The Shore of Closed-String Gravity: an Exact Unitarity Boundary for the Cheung–Hillman–Remmen Graviton Family

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Abstract

The level-truncation bootstrap of Cheung, Hillman and Remmen produces a one-parameter family of graviton amplitudes containing the Virasoro–Shapiro amplitude at $\lambda = 1$, with residues that are perfect squares; partial-wave positivity was known to bound λ from below for $D \geq 9$, numerically. We derive the boundary in closed form. The near-leading even trajectory obeys $a_{n,2n-4} \geq 0 \iff D \leq T_n(\lambda)$ with $T_n(\lambda) = \frac{3(2n-3)}{n(n-2)}\big(\lambda^2 + (2n-2)\lambda + 1\big) + 2n$, verified exactly for $n \leq 14$ and against exact-rational scans. On this trajectory the boundary is the envelope $\min_n T_n$ — and we conjecture (with the falsification battery reported below) that it is the boundary of the full family. The shore is born at $(D, \lambda) = (9, 0)$ — explaining the onset observed by CHR — passes *exactly* through the string at $D = 23$ (so pure Virasoro–Shapiro first loses positivity on this trajectory at $D = 24$, via the level-4 spin-4 wave, matching exact scans), and straightens onto the exact asymptote $D = (12 + 4\sqrt{3})\,\lambda$ with active level $n^* \simeq \sqrt{3}\,\lambda$. Doomed-cell predictions of the envelope are confirmed by direct evaluation. We conjecture the envelope is the complete boundary and report the falsification battery it has survived.

1 Setup

Family (CHR, PRD 111, 086034): $\mu(n) = \frac{n+\lambda-1}{\lambda}$, $R(n,t) = \big[\,((1+\lambda)/2 + \lambda t)_{n-1}/((1+3\lambda)/2)_{n-1}\big]^2$, massless external states ($s+t+u = 0$), $\lambda \geq 0$, VS at $\lambda = 1$. Partial waves: Gegenbauer coefficients at $s = \mu(n)$, $t = -\mu(n)(1-x)/2$; odd waves vanish by the $t \leftrightarrow u$ symmetry of the squared residue. All computations exact-rational; evaluator validated against CHR's reported behavior (D=4 positivity for all λ; onset of the lower bound at $D \geq 9$).

2 The trajectory law

Theorem 1 (Near-leading even trajectory)**.** *For the residues above, $\lambda > 0$ and $D > 3$,*

$$a_{n,2n-4} \;\geq\; 0 \quad\iff\quad D \;\leq\; T_n(\lambda), \qquad T_n(\lambda) = \frac{3(2n-3)}{n(n-2)}\big(\lambda^2 + (2n-2)\lambda + 1\big) + 2n. \tag{1}$$

Derivation. Write $\lambda\big(t - \xi(k)\big) = \frac{1}{2}\big[(n+\lambda-1)x + (2k-n)\big]$ at level n; the residue is the square of $Q(x) = \prod_{k=1}^{n-1} \lambda(t - \xi(k))$ up to a positive factor. The roots $x_k = (n-2k)/(n+\lambda-1)$ are symmetric under $x \to -x$, so Q has definite parity: odd partial waves vanish identically, and

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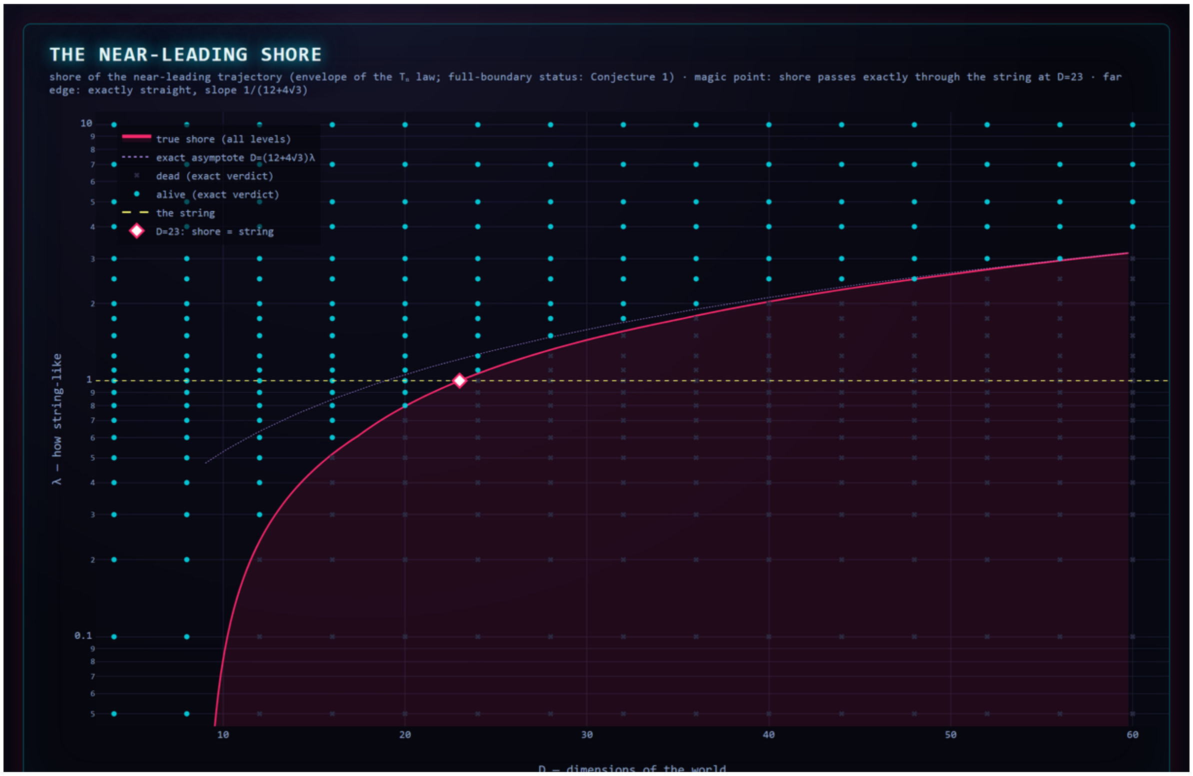

Figure 1: The boundary of the family: exact verdicts (cyan alive, grey dead; λ on a log scale), the analytic shore $\min_n T_n(\lambda)$, its straight asymptote, and the point $D = 23$ where the shore passes exactly through the Virasoro–Shapiro amplitude.

the subleading coefficient q_{n-2} of Q vanishes. Hence the x^{2n-4} coefficient of Q^2 is the *pure cross term* $2q_{n-1}q_{n-3} < 0$ — a naive term-by-term estimate would miss the edge entirely. Only x^{2n-2} and x^{2n-4} feed the Gegenbauer mode $\ell = 2n-4$ (parity and degree), with the exact projection ratio $\rho(\ell, D) = \frac{(\ell+1)(\ell+2)}{2(D+2\ell-1)}$; assembling the two contributions and removing positive factors yields T_n. Using $e_2(\text{roots}) = -\frac{n(n-1)(n-2)}{6(\lambda+n-1)^2}$ one obtains the equivalent closed form $T_n = \frac{3(2n-3)(\lambda+n-1)^2}{n(n-2)} - (4n-9)$.

Verification. The bracket was extracted independently by symbolic computation for every $n \leq 14$ and matches T_n identically (12/12); an independent adversarial review re-derived the theorem from scratch, and its falsification suite (published with the code) found no counterexample, including razor zeros at fractional dimensions $D^* = 271/4$ (n=5, λ=7/2) and $D^* = 57$ (n=6, λ=3) computed directly from the projection with symbolic D. Doomed-cell executions confirm predicted kill levels exactly (e.g. $(\lambda, D) = (3/2, 32)$ and $(2, 40)$: sign flip at $n = 4$ as predicted).

3 The envelope

The boundary implied by Theorem 1 is the envelope $\lambda_{\min}(D)$ defined by $D = \min_{n\geq 3} T_n(\lambda)$. Its structure:

Onset at $D = 9$. $T_n(0) = 2n + \frac{3(2n-3)}{n(n-2)}$ is minimized at $n = 3$ with $T_3(0) = 9$: the trajectory constrains nothing for $D \leq 9$ — reproducing, as a formula, the onset CHR observed numerically ($D \geq 9$).

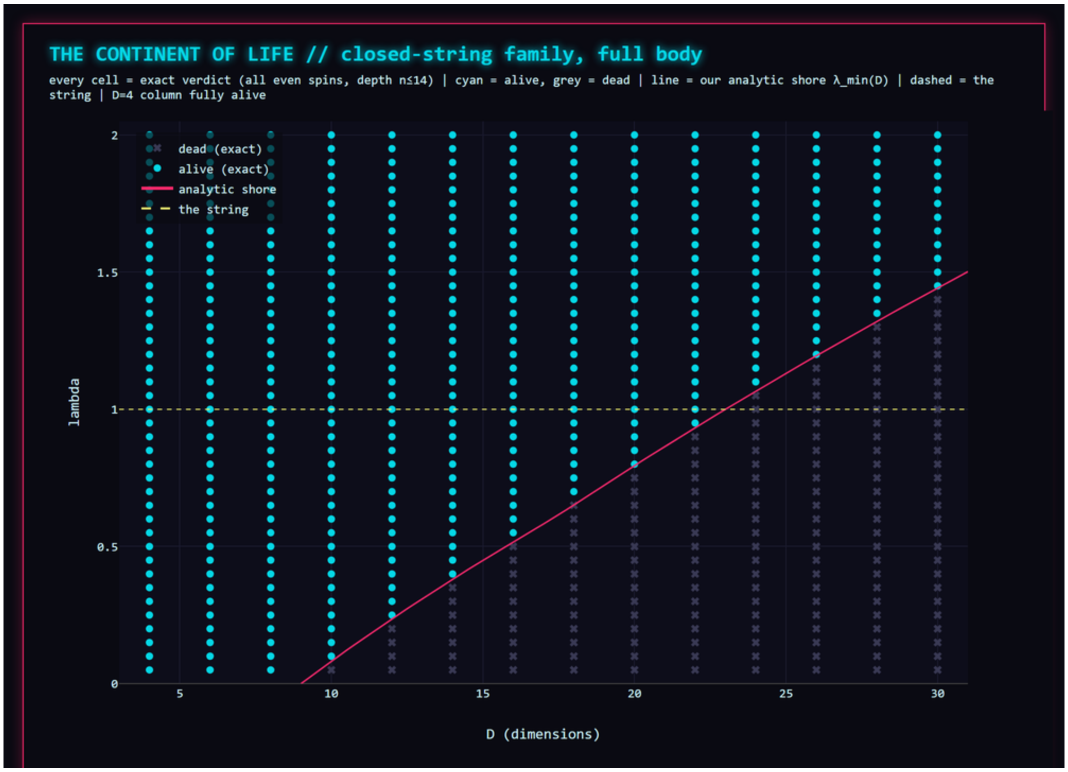

Figure 2: The allowed body at linear scale with the near-leading shore overlaid; every cell is an exact rational verdict (all even spins, depth 14).

The magic point. $T_n(1) = \frac{2(n^2+4n-9)}{n-2}$, whose integer minimum is 23, attained only at $n = 4$ (continuous minimum at $n = 2 + \sqrt{3}$). Hence $\lambda_{\min}(23) = 1$ *exactly*: the shore passes through the Virasoro–Shapiro point, and pure closed-string gravity first loses positivity on this trajectory at $D = 24$, through its level-4 spin-4 wave — matching the exact scan (first negative at $(n, \ell) = (4, 4)$ for $D = 24$, clean at $D = 22$ to depth 40).

The straight far edge. Minimizing T_n over continuous n at large λ gives the active level $n^* = \sqrt{3}\,\lambda$ and the exact asymptote $D = (12 + 4\sqrt{3})\lambda$; numerically $D/\lambda_{\min}$ rises $18.75 \to 18.92$ over $D = 60 \to 10^4$ against $12 + 4\sqrt{3} = 18.928\ldots$, with the residual $O(1/\lambda)$ oscillation caused by the discreteness of n^*.

4 Virasoro–Shapiro thresholds

For the string itself the theorem gives the threshold sequence $D_n(1) = 2(n^2 + 4n - 9)/(n - 2) = 24, 23, 24, \frac{51}{2}, \frac{136}{5}, \ldots$ — an exact refinement, on this trajectory, of the finite-level $D_{\text{crit}}(n)$ phenomenology known for Veneziano-type amplitudes. Because $D_n(1)$ grows linearly, the trajectory endangers VS only within a finite window of levels at any fixed D; the deep asymptotic bounds of the literature arise from other (low-spin) constraints and are complementary to the exact finite-level statement proven here.

5 Conjectured completeness and its battery

Conjecture 1. *A point (λ, D) of the family is consistent with partial-wave positivity at all levels iff $D \leq \min_{n\geq 3} T_n(\lambda)$.*

The conjecture is falsifiable: a single exact negative coefficient at a point with $D \leq \min_n T_n$ refutes it. Its current battery: 494/494 agreements with exact full-spin scans (all even $\ell \leq 2n-2$, depth 16, grid $\lambda \in [0.1, 10]$, $D \in [4, 40]$) with zero alarms of either type; two direct doomed-cell executions at the predicted level; and an independent adversarial review whose suite (seven attack batteries) found no counterexample. The reviewer-designated most plausible additional knife — the $\ell = 2n-6$ trajectory — has been hunted across stress zones hugging the conjectured boundary (34 values of λ, depth windows $[\min_n T_n - 3, \min_n T_n]$, $n \leq 14$): zero alarms; further trajectories and low-spin stress tests are running as standing batteries. In $D = 4$ the model predicts (and scans confirm, 40/40 to depth 14 plus the trivial bound $T_n \geq T_n(0) \geq 9 > 4$) that the entire family survives: within these assumptions, four-dimensional gravity is *not* forced to be string-like at any finite depth we can reach.

6 Discussion

Together with the companion open-string results, a graded picture emerges: in $D = 4$ the entire λ-family survives every test we can run — at this level of assumptions, four-dimensional graviton scattering is not forced to be string-like — while as D grows the shore rises and passes exactly through the Virasoro–Shapiro point at $D = 23$: the string becomes extremal precisely in the gravitational, high-dimensional setting. The method (top coefficients of residue polynomials through the partial-wave projection) transfers verbatim between families and is cheap: every number here was computed on a desktop in exact arithmetic. Natural next steps: closing Conjecture ef-conj:model; the planar analogues CHR construct; and mapping the surviving $D = 4$ alternatives to their effective-field-theory coefficients, where gravitational-wave phenomenology could in principle discriminate them from the string.

Reproducibility and AI disclosure

All claim-bearing computations use exact rational arithmetic; the evaluator, the falsification suites (the adversarial reviewer's seven-battery script included), all scan artifacts, and the append-only research log are released with this paper. This research was conducted by the author working with an AI research assistant (Claude, Anthropic); every derivation passed an independent adversarial review, and every number in this paper is regenerated by a single published battery script. During this work an implementation bug (a Pochhammer increment step, coinciding with the correct evaluator only at $\lambda = 1$) was found, disclosed and fixed in the public log; all results affected by it were voided and recomputed, and the $\lambda = 1$ results it could not affect were re-verified independently.

References

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